Test Case Generation with Hecate: To Infinity and Beyond!

This repository contains the necessary files to replicate a series of tests described in the *Test Case Generation with Hecate: To Infinity and Beyond!* paper. The tests involve using the Hecate tool for test case generation and Simulink models to evaluate system behavior against defined requirements.

Repository Folders

This repository contains two folders, i.e., one for each of the two analyzed case studies (`Drone_controller` and `eBike_controller`).

eBike_controller

The folder contains:

- two Simulink models (Buck and PWM)
- a script to run Hecate on them
- an initialization file containing model parameters.
- an Excel file containing the results of our experiments.
- a Python script performing the statistical tests between number of iterations and time, for the comparison between Uniform Random and Simulated Annealing.

Requirements

To run the script and be able to open the Simulink models, ensure the following software is installed:

- MATLAB version R2024a or newer and the following Add-Ons:
  - Simulink
  - Simulink Test
  - Simscape Electrical (Simscape)
  - Parallel Computing Toolbox
- Hecate (Follow the instructions written in the tool repo)

How to Install

In MATLAB, navigate to the Hecate repository and add the folders `src` and `staliro` to the active path. The *genpath* function takes also the subfolders.

```
addpath("src")
addpath(genpath("staliro"))
```

Then, add also the `eBike_controller` from this repository to the active path.

```
addpath("Replication_Package")
```

How to Run

After adding to the active path the necessary folders, navigate into the `eBike_controller` folder and run the following command to execute the test.

```
runTest;
```

Possible configurations

In the *runTest* script, the user can change:

- the *Simulink Model* changing the *modelName* variable (*Buck_model* or *PWM_model*)
- the *Test Sequence Scenario* changing the *hecateOpt.sequence_scenario* variable (6 available scenarios)
- the *Test Assessment Scenario* changing the *hecateOpt.assessment_scenario* variable (3 available scenarios)

When the user changes the Test Sequence Scenario or the Assessment Scenario, both must be activated from the Simulink Model by opening the *Test Sequence* Block and the *Test Assessment* Block and selecting the right scenario in the Tab Scenarios. The chosen scenario has a little thunderbolt icon.

Statistical test results

`Wilcoxon Rank Sum Test on average number of iterations {'statistic': 0.037904902178945175, 'p_value': 0.9697635044861096, 'reject_null': False, 'effect_size': -0.11923687194166115, 'confidence_interval': (-4.8031564689160655, 3.5013173884562923)}`

`Wilcoxon Rank Sum Test on times {'statistic': 0.16262867961000602, 'p_value': 0.8708108045360416, 'reject_null': False, 'effect_size': -0.24753480517182908, 'confidence_interval': (-4915.796179129435, 2388.6628457961033)}`

Drone_controller

The folder contains:

- a Simulink model of a drone controller with its Simulink project (`MinidroneCompetition.prj`)
- a script to run Hecate on the drone controller (`runTest_hecate.m`)
- a folder containing the results of our experiments (`Hecate_results`)

Requirements

To run the simulation and be able to open the Simulink model, ensure the following software is installed:

- MATLAB version R2024b or newer and the following Add-Ons:
  - Simulink
  - Stateflow
  - Aerospace Blockset
  - Aerospace Toolbox
  - Computer Vision Toolbox
  - Control System Toolbox
  - Global Optimization Toolbox
  - Image Processing Toolbox
  - MATLAB Coder
  - MATLAB Support for MinGW-w64 C/C++/Fortran Compiler
  - Optimization Toolbox
  - Parallel Computing Toolbox
  - Signal Processing Toolbox
  - Simulink 3D Animation
  - Simulink Support Package for Parrot Minidrones

How to Install

In MATLAB, open the *MinidroneCompetition.prj* project file. This action will initialize the project path, set up working folders, open the Quadcopter Model and the simulation.

How to Run

After the Simulink model is open, in MATLAB execute the file

```
runTest_hecate.m
```

To generate different tracks and test the chosen controller using Hecate.

The function used by Hecate to generate new paths is in the file `generate_path.m`.

Results

In the `Hecate_results` folder you can find:

- **`testSA_Drone.mat`** — contains the test results for the comparison with the drone (S-Taliro vs Hecate) using the SA algorithm
- **`testSA_Hecate.mat`** — contains the test results for the comparison with the e-Bike using the SA algorithm
- **`testUR_Hecate.mat`** — contains the test results for the comparison with the e-Bike using the UR algorithm

Possible configurations

Inside the *runTest_hecate.m* file, the user can select different S-Taliro options:

- `opt.optimization_solver`: optimization algorithm (UR_Taliro, SA_Taliro);
- `opt.runs`: number of runs;
- `opt.optim_params.n_tests`: number of tests per each run;
- `opt.save_intermediate_results`: set 1 to save intermediate results during the experiments, otherwise 0.

The user can also select the parameters to generate different tracks:

- `controller_selector`: the controller version under test (1,2,3);
- `angle_threshold`: the minimum valid angle;
- `distance_threshold`: the minimum distance between two lines;
- `allow_interstions`: set 1 to allow intersections between lines.

Contributors

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